Accepted Manuscript

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PII: S0304-3878(18)30437-1

DOI: 10.1016/j.jdeveco.2018.07.002

Reference: DEVEC 2261

To appear in: *Journal of Development Economics*

Received Date: 20 May 2016

Revised Date: 1 June 2018

Accepted Date: 3 July 2018

Please cite this article as: Keats, A., Women's schooling, fertility, and child health outcomes: Evidence from Uganda's free primary education program, *Journal of Development Economics* (2018), doi: 10.1016/j.jdeveco.2018.07.002.

Women's Schooling, Fertility, and Child Health Outcomes: Evidence from Uganda's Free Primary Education Program

Abstract

This paper examines the role of womens education on both fertility and child health in Uganda. To identify causal effects, I exploit the timing of a national reform that eliminated primary school fees in 1997 to implement a regression discontinuity design. At the cutoff, the reform increased educational attainment by nearly one year on average, with impacts across all grade levels through the end of secondary school. Women with more schooling both delay and reduce overall fertility, increase early child health investments, and have less chronically malnourished children. In terms of mechanisms, women with additional schooling do not abstain more from sex as adolescents, but they are more likely to have used contraceptives before a first pregnancy and they delay marriage. Other downstream effects include improved employment outcomes and greater wealth.

Keywords: Education, Fertility, Child Health

JEL: O12, I12, I25, I38

1. Introduction

Neoclassical models of fertility predict that increased female education induces a trade-off between child quantity and child quality (Becker (1960) and Becker and Lewis (1973)). Empirically, the correlations between female schooling, lower fertility, and improved child health and human capital are well known (Cochrane (1979), Strauss and Thomas (1995), and Grossman (2006)), and serve as the basis for the focus on women's schooling in developing countries as a means to address a broad set of social objectives. Indeed, the World Bank has said "there is no investment more effective for achieving development goals than educating girls." Yet there is no direct evidence that rising female education *both* reduces fertility and improves child investments and outcomes.

While a growing body of experimental and quasi-experimental studies show that education delays fertility, evidence is mixed on whether overall fertility declines as well. Causal evidence that education reduces teenage pregnancies has been found in several developing countries, including Nigeria (Osili and Long (2008)), Malawi (Baird et al. (2010)), Kenya (Ferré (2009), Ozier (2018), and Duflo et al. (2015)), and Indonesia (Breierova and Duflo (2004)), as well as in more developed countries, including the U.S. and Norway (Black et al. (2008) and Monstad et al. (2008)) and Israel (Lavy and Zablotsky (2015)).[1] However, among the few studies that are able to explore whether schooling reduces overall fertility, the results are less clear. While some find that education does lead to fewer total children born (Osili and Long (2008) and Lavy and Zablotsky (2015)), there is also evidence that more educated women who delay fertility in adolescence eventually catch-up to less educated peers (Breierova and Duflo (2004) and Monstad et al. (2008)).

In terms of child health, there are relatively fewer studies that rigorously examine the connection between mothers' education and child outcomes and the

[1]In contrast, McCrary and Royer (2011) find no effect of education on teenage pregnancy among drop outs in the U.S.

evidence from these studies is also mixed. Breierova and Duflo (2004), Chou et al. (2010), and Currie and Moretti (2003) find positive effects of parental education on child health in Indonesia, Taiwan, and the U.S., respectively. On the other hand, McCrary and Royer (2011) and Lindeboom et al. (2009) find small or insignificant effects on child health in the U.S. and U.K., respectively. Only Breierova and Duflo (2004) and McCrary and Royer (2011) examine both fertility and child health outcomes simultaneously. But while both find positive effects of education on child health, because there are no effects on overall fertility, neither provides evidence in support of the quality-quantity tradeoff predicted by neoclassical fertility models.

This paper exploits the timing of Uganda's Universal Primary Education (UPE) program, which eliminated primary school fees beginning in 1997, to contribute new evidence on the impacts of women's schooling on both fertility and child health. In removing fees, Uganda is not alone – nearly half of the countries in Sub-Saharan Africa have introduced (or re-introduced) free primary education policies since 1994. Yet there are few studies that measure long-run educational and social impacts of these efforts. Deininger (2003), Nishimura et al. (2008), and Grogan (2008) find Uganda's UPE reform increased primary school access and enrollment, especially for girls, but do not measure overall attainment. Studying Kenya's 2003 Free Primary Education program, Bold et al. (2010), Lucas and Mbiti (2012a), and Lucas and Mbiti (2012b) similarly show gains in access and primary school completion. Only Osili and Long (2008) measure impacts on overall educational attainment and fertility, but they do not measure child health outcomes.

Using data from Demographic and Health Surveys (DHS), I show the Uganda UPE reform led to an immediate surge in primary school enrollment and, for girls, differential impacts on overall educational attainment across birth cohorts. In particular, there is a discontinuous jump in completed schooling of 0.72 years for the cohort of women who were 14 years old or younger in 1997 (born in

1983 or later).[2] The location of this discontinuity is consistent with school starting ages and the length of primary school. Women who were 14 years old or younger in 1997 were disproportionately more likely to be attending primary school and thus benefit from the reform, while women from older cohorts were not (either because they had moved on to secondary school or had dropped out). The estimated discontinuity is robust across the several datasets used in this paper both individually and combined, and stable across a range of specifications, bandwidths, and controls.[3] Since women born on either side of the cutoff share similar background characteristics, and since year of birth is exogenous to them, the implementation of the UPE reform provides variation in educational attainment across age cohorts that is "as good as randomized" (Lee (2008)).

Remarkably, the gains in schooling as a consequence of the reform appear at every grade level, including through the end of secondary school (where school fees remained in place). While effects are largest for the final few years of primary school, affected women are also discontinuously more likely to have completed primary school, attended some secondary school, and to have graduated from secondary school. This suggests that the effects of the UPE program extended to women from all points in the skills distribution, i.e. that primary school fees prevent even high ability women from completing more schooling. It also suggests that the LATE results presented in this paper are closer to the average treatment effect of increasing female schooling.

There are four sets of main results. First, women who get more education in Uganda both delay the onset of childbearing and have fewer children overall.

[2] As discussed in Section 3, I find no discontinuity in education for men using DHS data from 2000/01, 2006 and 2011, nor in other cross-sectional data. See also Appendix Figure A2 and Appendix Table A1.

[3] Although not used in the analysis in this paper, a similar discontinuity in educational achievement can be found in the 2002 Uganda Census, the 2008 Uganda National Service Delivery Survey, and the 2013 Uganda Living Standards Measurement Study survey. See Appendix Table A1.

An additional year of schooling reduces the likelihood of having a first birth by each age from 16-20 years old by about 11 percent on average (by age 21, women affected by the reform are just as likely to have had a first birth as women who were not affected). This effect is similar in sign and magnitude to other experimental and quasi-experimental studies from sub-Saharan Africa (including Ferré (2009), Duflo et al. (2015), and Ozier (2018) in Kenya, Baird et al. (2010) in Malawi, and Osili and Long (2008) in Nigeria).

I also find that education reduces overall fertility at each age through to 25 years old. In fact, the results show that the difference in the number of children born grows in absolute value as women age, suggesting both that the effect of schooling on fertility persists beyond the time when most women have left school and that women with more schooling do not "catch-up" later on. This is further supported by the fact that, among women with multiple births, the average spacing between births is greater for those women affected by the reform.

Second, this paper examines the causal relationship between mothers' education and child health outcomes, which has not been convincingly demonstrated thus far in Sub-Saharan Africa. Since women affected by the reform have fewer children at any age, I focus on health outcomes of first-born children. I find that greater female education leads to increased investments in first-born children, but with mixed results in terms of child health outcomes. First-born children of women with additional schooling are twice as likely to be born with a trained health provider present and are about 12 percent more likely to receive vaccinations by 1 year of age. They are also less likely to display signs of chronic malnutrition. For each additional year of mother's schooling, first-born children are 0.43 standard deviations taller for their age and 37 percent less likely to be stunted. However, I find no differences in mortality rates (both in infants and children under 5) for first-born children of women on either side of the cutoff.

Third, I explore potential channels for the fertility and child health results and present evidence on other downstream impacts. Many potential mechanisms have been suggested to explain how education could impact decisions about fertility and investments in child health (see McCrary and Royer (2011) for a review). Because most women in the sample have begun having children before the survey date, I focus on outcomes contained in the DHS that were plausibly fixed before the onset of childbearing to sharpen the causal connection. These include age at first intercourse, contraception use before first birth, age at marriage, and education of husband.

I find that while education does not delay the onset of sexual activity, women with more schooling do delay marriage, particularly in their adolescent years. In addition, women with more schooling are 36 percent more likely to have used contraception before the birth of their first child (an increase of 6.5 percentage points off of a base of 18 percent). This latter result is consistent with De Walque (2007), who finds that women in Uganda with more schooling are more likely to respond educational campaigns promoting safe sex practices, and Duflo et al. (2015), who find that while increased education in Kenya does not reduce the incidence of sexual activity, it does increase the adoption of strategies to avoid pregnancies. There is, however, no evidence of positive assortative mating among the women with more schooling who do marry.

Women affected by the reform further demonstrate other downstream impacts that, while potential channels themselves, are also potentially endogenous responses to the main fertility and child health results. In particular, women with an additional year of schooling are 9.2 percent more likely to have worked in the last year (a result which mirrors the extensive margin effects in Duflo (2001)), and, conditional on working, they are 33 percent more likely to be employed by another person (rather than self-employed) and 13 percent more likely to be paid in cash (rather than in kind or not at all). Perhaps as a consequence, women with more schooling also generally better off. They score significantly higher on a composite wealth index of household assets (including ownership of televisions, radios, and refrigerators, as well as the type of construction used in the home, and access to clean water and sanitation).

Finally, I use a difference-in-difference-in-differences approach to shed light on whether the main results are being driven in fact by an education effect rather than by an income effect. An income effect is plausible given that there are some women (typically from better-off households) who would have finished primary school in the absence of the UPE reform, and for whom, therefore, the elimination of primary school fees served principally as a household income transfer. The triple differences framework compares outcomes of relatively wealthier and poorer affected and un-affected women before and after the reform. This analysis provides evidence that the main effects are indeed driven by women who are induced by UPE to get more schooling (i.e. the less well off), and not by household income gains accruing to those women who would have attended primary school in any event.

The remainder of this paper proceeds as follows. Section 2 describes the data that I use in the analysis. Section 3 provides information on the schooling system in Uganda, and describes the 1997 Universal Primary Education reform and it's immediate effects on enrollment and grade achievement. Section 4 presents the empirical strategy. Section 5 presents the results. Finally, Section 6 concludes.

2. Data

The data used in this paper come from the 1995, 2000/01, 2006, and 2011 Demographic and Health Surveys (DHS), and the 2009 Malaria Indicator Survey (MIS). Each survey round features a representative sample of women aged 15-49 and typically includes information about education and employment, knowledge and use of family planning and other health care, as well as complete birth histories, and, for children born in the 5 years preceding the survey, child health outcomes.[4] The DHS data also include information on men and children aged 5-25 who still live at home.

For the main results, I pool the data to increase sample size and reduce noise.[5] As described in section 5.1, data from each survey round are restricted

[4] One exception is that the 2009 MIS has only limited information on child health outcomes.
[5] Results do not change significantly if the data is not pooled.

to include only women from the eight birth cohorts on either side of the cutoff (i.e. women born between 1975-1990). Further, I exclude women from any cohort who were 15-18 years old at the time they were surveyed since many of these women were still in school and therefore it is unclear what their ultimate educational attainment would be.[6] Given these restrictions, the pooled sample draws from the 2000/01, 2006, and 2011 DHS and 2009 MIS. Table 1 provides summary statistics, by survey round, for this sample.

I use a slightly modified approach for the child health results. Data are again pooled and restricted to women from cohorts born between 1975-1990. However, because this sample consists only of women who have borne children, and because mothers rarely return to school in Uganda, this sample does not drop women less than 19 years old at the time of the survey. Further details on how this sample is created are discussed more thoroughly in Section 5.6.

Finally, I combine other data to explore effects around the time of the UPE reform and to compare the impact of the reform across genders. I use the 1995 and 2000/01 DHS household surveys to chart attendance and grade completion of women and men born around the cutoff year when they were still of school age. This data also provides background information on the wealth and education levels of the parents of people who were born around the cutoff used to confirm the smoothness assumption underlying the validity of the regression discontinuity design. The 1995 and 2000/01 DHS women's data are also used for the difference-in-difference-in-differences estimation described in section 5.5. Finally, since the DHS have data on men's educational outcomes as well, I also combine the 2000, 2006 and 2011 men's surveys to measure the impact of free primary education for men (again restricting the data to cohorts born between 1975-1990 for comparability).

Notes explaining which data are used for each result are included throughout this text and in the tables and figures.

[6] Estimates of treatment effects at the discontinuity are not sensitive to small adjustments to this restriction in either direction.

3. Primary Schooling in Uganda and UPE

The public education system in Uganda consists of 7 years of primary school, 4 years of lower secondary school, and 2 years of upper secondary school. The academic year has 3 terms beginning in February and ending in December. Officially the school starting age is 6, but according to the 1995 DHS data, on the eve of the reform most children began school closer to age 8 (just 16 percent start later). Overall, academic achievement is low, with clear disparities across genders: boys aged 6-16 are more likely to attend primary school (64 percent) than girls of the same ages (58 percent); just 20 percent of women and 30 percent of men attend any secondary school; and on average women complete 4.7 years of schooling while men complete 6.5 years.

3.1. Primary school attendance patterns before UPE

Prior to the UPE reform, most children aged 15 and older had either dropped out of primary school or moved on to secondary school. Given the low levels of secondary school enrollment, most of these children are dropouts. According to the 1995 DHS, among girls aged 15-20 who had dropped out of primary school before the reform, 80 percent cited the cost of schooling as the main reason for stopping.[7] Figure 1 shows the fraction of girls by age group with less than a primary school education who had not dropped out by 1995. Consistent with the *de facto* school starting age of 8 prior to the UPE reform, and the 7 years required to complete primary school, there is a steep decline at age 15. Whereas at age 14 more than 67 percent of girls are still enrolled in primary school, this figure drops to 44 percent for girls aged 15. In contrast, for boys, the gap in primary school attendance between 14 and 15 year olds is less than 10 percentage points (74 percent vs. 65 percent). Although it was possible for children to re-enter primary school after dropping out, this abrupt change in the likelihood of being in primary school at the time of the reform provides plausible support for the differential effects of the UPE program on girls by age, particularly around

[7] Similar information is not available for males.

the ages of 14 and 15 (i.e. between the 1982 and 1983 cohorts), estimated in this paper.

3.2. The UPE reform

In the 1997 school year, the government of Uganda introduced the UPE program, in which it pledged to eliminate primary school fees, among other reforms.[8] Previously, parents paid 5,000 Ugandan shillings (Ush) for each child in primary grades 1-3 and 8,100 Ush for children in grades 4-7 (approximately 5USD and 8USD in 1997 dollars, respectively).[9] Parents remained responsible for other school-related costs, such as uniforms, books and supplies, and transport. Parents were notified about the reform starting approximately 2 months prior to the 1997 school year and there is evidence that by 2001 nearly all parents had heard of the reform (97 percent) and understood that they were no longer responsible for school fees (96 percent) (Uganda Bureau of Statistics (UBOS) and Macro (2001)). The reform also introduced measures aimed at reducing the rate of capture of educational funds by local politicians and administrators, and thus may have effectively increased government assistance at the school level.[10]

The introduction of UPE resulted in large increases in primary school attendance for both boys and girls and a significant jump in educational attainment for girls who were aged 14 and younger at the time of the reform.[11] Appendix Figure A1 presents primary school attendance rates before and after the reform by age for both girls and boys aged 5-25 using data from the 1995 and 2001 DHS, respectively. On average, primary school attendance increased by 37 percentage points for 5-year-olds and by 12 percentage points for children between

[8]Although originally the government planned to pay for up to 4 children per household (2 of which had to be girls if daughters were present), in practice all children were covered. Other reforms included the provision of classrooms and desks, increased teacher training, and a revision and update of the curriculum.

[9]In 1997 GDP per capita was 283USD.

[10]See Reinikka and Svensson (2005) for details.

[11]There is also evidence that the surge in school attendance sharply increased student-teacher ratios and strained other resources causing a drop in school quality Grogan (2008).

the ages of 6 and 15 (the principal ages for primary schooling).[12] Gains were larger for girls than for boys, particularly for girls aged 13-15, suggesting that school fees were a larger constraint on girls than boys at these ages. These disproportionate increases effectively closed the gender gap in primary school attendance which existed prior to the reform. Gender gaps in overall educational attainment and in the probability of attending secondary school closed as well. Appendix Figure A2 presents average grade completion, by birth-year cohort, for men using the 2000/01, 2006, and 2011 DHS. There does not appear to be any change from trend in educational attainment among men.[13] This is perhaps not too surprising given that the average male born in cohorts likely to be unaffected by UPE (those who were 15 and older in 1997) already achieved 7 years of schooling prior to the reform, and it supports the notion that school fees were not a large constraint on primary schooling for boys.

Focusing on women's educational outcomes, I pool data from the DHS surveys from 2000/01, 2006, and 2011, along with the 2009 MIS, to plot educational attainment by birth-year cohort in Figure 2. In addition, I plot a linear trend line estimated separately for women born between 1950-1982 and for women born between 1983-1990. The figure shows a discontinuous jump in average educational attainment at the cutoff of approximately .70 more years of schooling. Moreover, women born in 1983 and after increased educational attainment at nearly all grade levels. Figure 3 plots the reverse CDF of the highest grade level achieved for 8 birth year cohorts on either side of the cutoff.[14] There is little change in the educational attainment distribution for women born between 1975 and 1982, but for women born in 1983 and onwards there is a clear break with higher educational attainment in each grade through to the end of upper secondary school. This is somewhat surprising given that secondary school fees

[12]See Grogan (2008) for a discussion of the impact of UPE on school starting age.

[13]I find no effect of the UPE reform on male education attainment in other data sets as well. See Appendix Table A1.

[14]I use a limited set of cohorts to make the figure easier to view. The main implications are nearly identical if all cohorts are included.

remained in place. It also suggests that primary school fees prevented even high ability women from continuing their education. The abrupt change in educational attainment at all schooling levels at the cutoff forms the basis for the empirical strategy used in this paper.

4. Empirical Strategy

The goal of this paper is to examine the relationship between women's education and a range of fertility and child health outcomes. However, estimation of this relationship using OLS is likely to produce biased estimates. Estimates may be biased upward due to unobserved factors (such as innate ability) that affect both educational attainment and the outcomes of interest, or biased downward because of measurement error in education levels. In order to mitigate these issues I take advantage of the timing of the UPE reform to implement a regression discontinuity design. As discussed in Section 3, the introduction of free primary education had a disproportionate effect on women born in 1983 and after, who were more likely to still be attending primary school in 1997 when the policy was put in place. Since the change in treatment (attending an additional year of primary school) jumps by less than 1 at the cutoff birth year, this is a fuzzy regression discontinuity design (Imbens and Lemieux, 2008).

Formally, in the fuzzy setting the causal effect of an additional year of schooling is defined as the difference in the outcome from a regression on the treatment-determining variable (birth year) divided by the difference in the treatment (an additional year of primary schooling) from a regression on the treatment-determining variable, both evaluated at the cutoff value, c:

$$\tau_{FRD} = \frac{\lim_{x\downarrow c} \mathbb{E}\left[Y|Birthyear = x\right] - \lim_{x\uparrow c} \mathbb{E}\left[Y|Birthyear = x\right]}{\lim_{x\downarrow c} \mathbb{E}\left[Educ|Birthyear = x\right] - \lim_{x\uparrow c} \mathbb{E}\left[Educ|Birthyear = x\right]} \quad (1)$$

As Hahn et al. (2001) note, this is numerically equivalent to the 2SLS estimator β_{FRD} with the system of equations:

$$Y = \alpha + \beta_{FRD}\widehat{Educ} + f(Birthyear - c) + \varepsilon \quad (2)$$

$$Educ = \gamma + \delta Treat + g(Birthyear - c) + \nu \quad (3)$$

where $Treat = \mathbb{1}\left[Birthyear \geq c\right]$, provided that the order of the polynomial and the bandwidth (in the case of local linear regression) are the same in both first and second stages. Restricting $f(\cdot)$ and $g(\cdot)$ to be of the same order allows standard errors to be computed as in 2SLS.

In order to determine the order of the polynomial functions $f(\cdot)$ and $g(\cdot)$ and the bandwidth (the data window), I follow Lee and Card (2008) and Lee and Lemieux (2010). I start by considering a piece-wise linear specification in year of birth, allowing the slope to vary on either side of the cutoff. I conduct a series of tests to determine the optimal bandwidth and, at each bandwidth, the appropriateness of the piece-wise linear specification against more flexible functional forms. The first such test is a modified leave-one-out cross-validation procedure (Imbens and Lemieux (2008)). The cross-validation function declines rapidly once the window is opened to include 4 birth-year cohorts and remains relatively stable until after 8 cohorts are included, when it gets slightly worse (Appendix Figure A3). This suggests a window with 4 to 8 cohorts on either side of the cutoff. At each bandwidth I then run additional regressions with higher order piece-wise polynomial controls (quadratic, cubic, and quartic). Using Akaike's information criterion I test to see which specification best predicts the data as the window gets larger. This test confirms the local linear specification for bandwidths between 4 and 8 years. I also add a set of birth-year cohort dummies to the local linear specification and test the joint significance of these dummies. They are not jointly significant, again confirming that the restricted model fits the data well. This procedure also provides a falsification test of the regression discontinuity design: joint significance would have indicated the presence of discontinuities at points other than the cutoff. Finally, in Figure A4, I plot the estimated coefficient of the discontinuity at each bandwidth. Starting at a bandwidth of 4 years, the coefficient is significant beyond the 5 percent level and remains essentially unchanged through 8 years. Thus, in the remainder of this paper I use a local linear specification and a bandwidth of 8 years on either side of the cutoff.

The fuzzy regression discontinuity design provides valid estimates under the assumptions that: 1) individuals are unable to manipulate treatment status, 2) covariates that affect both the schooling decision and outcomes of interest vary smoothly across the cutoff, and 3) outcomes at the cutoff change solely as a result of women's additional schooling. I test the smoothness assumption by examining respondents' mothers' education, literacy levels, geographic location (living in an urban area), and wealth. Since the DHS surveys do not contain information on respondents' mothers' outcomes, I use the 1995 and 2000/01 DHS to create a sample of women who had children within the data window (between 1975 and 1990). These are not necessarily the mothers of the women in the main sample used for the results in this paper, but since the DHS data is representative, they are plausibly representative of their mothers. As shown in Figures A5-A8, the characteristics of mothers who had children between 1975 and 1990 show no signs of discontinuity around the cutoff, providing supportive evidence of the smoothness assumption.

The treatment effect is a local average treatment effect. Specifically, it is the effect of schooling on an outcome Y for compliers (those that would go to school for an additional year if school fees were eliminated, but who would not if fees remained in place) measured at the cutoff. However, since the UPE reform resulted in a surge of enrollment and gains in schooling for women at nearly all education levels, this suggests that the effects measured in this paper are not being driven by outliers in the education or ability distributions.

5. Results

5.1. Effects of UPE on education

The results of the first stage estimation (the jump in the years of schooling at the discontinuity) are presented in Table 2.[15] Column 1 shows the effect on educational attainment. At the discontinuity, women exposed to UPE complete

[15]See also Appendix Table A1 for evidence of the discontinuity in other nationally representative samples not used in this paper.

.72 more years of schooling relative to the cohort to the left of the cutoff. Just prior to the reform, women completed 5.63 grades on average, while after they completed 6.35 grades on average. Further, the interaction of the treatment indicator with the control for year of birth is positive and statistically significant, which is consistent with the notion that longer exposure to free primary schooling has even larger impacts on completed schooling. The discontinuity for the full sample is presented graphically in Figure 5.

As discussed in Section 3 and shown in Figure 3, the introduction of UPE increased educational attainment at all grade levels for women on the right-hand side of the discontinuity. Columns 2-4 of Table 2 show the impact of the reform on primary school completion, the probability of transitioning to secondary school, and secondary school completion, respectively. Women who experienced at least one year of free primary schooling were 5.7 percentage points more likely to complete primary school (a 15 percent increase relative to the 39 percent of women who completed primary school in the preceding birth cohort). The reform also increased the transition rate to secondary school and the probability of secondary school completion, a somewhat surprising result given that secondary school fees (which are typically larger than primary school fees) remained in place. Exposure to free primary schooling increased secondary school going by 17 percent (a 4.8 percentage point gain relative to a base of 29 percent), and boosted the probability secondary school completion by 30 percent (an increase of 2.8 percentage points off a base of 9 percent).

Finally, column 5 presents impacts on literacy. This is motivated by the fact that the sudden and massive increase in primary school attendance triggered by the reform may have also reduced school quality. If so, gains in educational attainment would not necessarily indicate increased learning. I find that – at least along the dimension of reading ability – this is not the case. As part of the DHS survey, women were shown a card with a simple sentence written in their native language and asked to read it out loud. I classify women as literate if they could read the entire sentence. At the cutoff, women who were exposed to at least one year of free primary schooling are approximately 8 percent more likely to be literate relative to women in the cohort to the left of the cutoff. Thus additional educational attainment does also correspond to increase learning.

5.2. Fertility outcomes

Education causes women to delay the onset of fertility. In Uganda, given traditionally low levels of schooling and the lack of many employment opportunities, women tend to start child-bearing early. The average age at first birth in the sample is 18. Approximately 25 percent of women begin having children at 16, and by age 21 more than 75 percent have had at least one child. Figure 5 plots estimates from separate regressions of the probability of having a first birth before age x on an indicator for being in birth cohorts to the right of the cutoff and piecewise linear controls.[16] Women who obtain additional schooling are less likely to have had their first birth at each age through 20. The effect is largest between the ages of 16-19, when a woman pursuing her education without interruption would be in secondary school. At age 16, education reduces the likelihood of a first birth by 6 percentage points (a 25 percent reduction relative to mean); at age 19 an additional year of schooling lowers this likelihood by 5 percentage points (an 8 percent reduction). In contrast, by age 21 treated women begin to be just as likely to have started child-bearing as women who were not exposed to the program. These results are not inconsistent with an "incarceration effect" in which women who are still in school have less incentive or opportunity to have a child (Black et al. (2008)).

However, the total number of children born is also declining with education, and this does not appear to be a purely mechanical effect caused by later starting age. The DHS contain data on all births, allowing a comparison of how many children women on either side of the cutoff have had at each age of their lives. Under the "incarceration effect" scenario treated women delay age at first birth, which, mechanically, would create a gap at the cutoff in the number of children

[16] Each regression is conditional on respondents having obtained at least age x by the date of the survey.

born during a woman's teenage years. However, this gap should eventually become flat and may shrink as treated women leave school and start having children. The data shows the opposite pattern: the gap in the total number of children born is increasing at higher ages. Figure 6 plots the IV estimates of the effect of schooling at each age through age 25. The estimates come from separate regressions in which the sample is limited to women age x and older, causing reductions in sample size after age 19. Between the ages of 16-18 an additional year of schooling leads women to have approximately .1 fewer children. But by the time women are 20 years old, this gap has increased to .16 fewer children, and by age 25 the gap is .24 fewer births.

A similar effect can be found by looking at the difference in the total number of children born at the cutoff using, sequentially, the 2006, 2009, and 2011 DHS data (Table 3). In 2006, when women to the right of the cutoff were 23 years old, an additional year of schooling decreases the number of children born by .09 children. Three years later in 2009, however, the difference in total children born increases to .36. Finally, using the 2011 DHS data when women to the right of the cutoff are 28 years old, this gap has grown to .52 fewer children (but is no longer statistically significant). Finally, column 4 of Table 3 examines the time between births among women with multiple children. On average, the spacing between births increases by 1.4 months for each year of female schooling. Taken together, these results suggest that women who get more education not only delay fertility but also are on a path to having fewer children overall (unless, of course, they continue child-bearing at ages older than women with less schooling).

5.3. Child health outcomes

5.3.1. Child health outcomes sample

So far the evidence presented in this paper suggests that increasing education leads women to have fewer children. As women limit the quantity of their offspring, an important question is whether they also increase the quality of the children they do have. In the present context, however, there are two challenges to exploring the effects of women's education on child outcomes. The first issue is the potential for differential selection given that women affected by the reform reduce their fertility. The second challenge is a data constraint: DHS data on child health inputs and outcomes is limited to children born only in the 5 years prior to a survey.

Ideally, to look at impacts of additional schooling on early child health investments and health one would need data on the entire, completed birth histories and child health outcomes of women affected and unaffected by the UPE reform. In this ideal sample there would be no differential selection.[17] There would be differences in total fertility, which could potentially be handled econometrically. However, it is not possible to create this ideal sample because women (particularly in the cohorts affected by the policy) are not old enough in the DHS surveys to have completed their fertility.

One option to deal with the fact that women's birth histories are incomplete would be to look at impacts of additional schooling on only the first-born child. In this case, to be entirely sure to avoid differential selection, one would need to have surveys of women who had at least attained the maximum age at which women have a first child (since education also delays the onset of fertility). In the DHS data, a few women have their first child in their early 30s. So, in order to have a sample of all first births and thus entirely avoid differential selection issues, one would need surveys of women who were at least in their early 30s. Yet, in the 2011 DHS, women to the right of the cutoff are at most 28 years old.

A second-best option, which I pursue here, is to exploit the fact that by age 21 (and beyond) treated women at the cutoff are just as likely to have had their first child compared to women to the left of the threshold (see Figure 5). Under the assumption that the treatment effect on the timing decision to begin fertility is monotonic – essentially just shifting the age at first birth upward, but preserving the underlying ordering among women of age at first birth based on

[17] That is, unless the reform causes women to bear no children at all. However, in the DHS data there is no evidence that education affects the probability of ever having a child.

individual characteristics – there should be no differential selection at the cutoff in the set of women who have had their first child by age 21.

Table 4 provides support for the validity of this approach. The first question addressed is whether there is evidence of differential selection at the cutoff into the sample of women who had their first child between the ages of 15 and 21. In order to test this, I first restrict the data to women who were at least 22 years old at the time of the survey to be able to correctly measure the probability of a first birth in the 15-21 year range by cohort. I also exclude the 2009 MIS since it does not contain data on most child health inputs or outcomes. Column 1 presents the first stage effect of UPE on schooling for this restricted sample. Reassuringly, these results look essentially identical to the first stage in the main sample (column 1 of Table 2). Thus, restricting the data to women 22 and older from each cohort, and dropping the 2009 MIS data, has minimal implications for the findings that follow.

Using this restricted data, column 2 explores the question of differential selection into the sample of women who had a first birth between the ages of 15-21. The dependent variable is now an indicator for whether a woman had her first child in this age range. There is no evidence of differential selection at the cutoff. While there is a downward trend in the probability of having a first birth between ages 15-21 by birth cohort, this trend is smooth at the cutoff and its slope is not altered for women born in cohorts after the cutoff.

Next, column 3 presents the first stage for the subset of women from the restricted data who also had their first child between the ages of 15-21. The first stage effect of UPE on schooling is nearly identical to the first stage presented in column 1 (as well as Table 2 column 1). Exposure to at least one year of free primary education increases schooling discontinuously at the cutoff by approximately .70 years, even for women who began childbearing between 15 and 21. However, while the results in column 2 suggest no differential selection, comparing columns 1 and 3 there does appear to be negative selection on education for this entire subgroup. On average, women who had their first child between the ages of 15-21 have 1 year less schooling compared to women who did not (5.72 compared to 4.74).

Although there is negative selection overall for this group, these women are representative of a large subset of the population. In the restricted data, nearly 80 percent of women aged 22 or older have begun child bearing by age 21, and 70 percent have their first child between ages 15-21.[18] They are also arguably the most policy-relevant subset of the population: namely, women who begin child bearing in adolescence and early adulthood. Nevertheless, in addition to the negative selection on education, women from this group are likely negatively selected on unobservables as well, so any differences in child outcomes presented below are likely lower bounds on the effects of education on average child health.

Columns 4 and 5 of Table 4 address the second issue in creating the child health input and outcome sample: data are limited to the children born in the 5 years preceding a DHS survey. I am interested in the sample of women who had their first child when they were 15-21 years old *and* who had this birth within 5 years of the DHS survey. I therefore now restrict the data to women who were 15-26 at the time of the survey since at these ages they are able to report on any child outcomes for births occurring in the last 5 years (when they were 15-21 years old). Given this new restriction, I now also include the 1995 DHS (along with the 2000/01, 2006, and 2011 DHS as before) to be able to include women from each of the eight cohorts on either side of the cutoff.

Column 4 checks for differential selection in this sample. The dependent variable is an indicator for whether a woman had her first child when she was 15-21 years old *and* this birth occurred within 5 years of the survey. As before, there is no evidence of differential selection at the cutoff.[19] Finally, column 5

[18]Of those women that do not have their first child in this range, roughly 7 percent have a first child before age 15 and 22 percent after age 21.

[19]Note, it is possible to further allay questions about differential selection by expanding the range further (from births between 15-21 to, say, 14-22 or beyond), but this would require excluding cohorts at the extremities of the sample. For example, in the 1995 DHS women in the oldest cohort (born in 1975) are 20 years old, and therefore do not report on outcomes for any children they may have had at 14. At the other extreme, in the 2011 DHS women

presents the first stage for this sample. Reassuringly, it is essentially identical to the first stage presented in column 3. Figure 7 graphically reproduces this first-born child sample first stage and compares it to the first stage among all women aged 15-26, regardless of whether they had a first child within the preceding five years of the survey.

5.3.2. Child health outcomes results

Turning now to the results, I first consider inputs to infant health (Table 5). While schooling has no impact on whether a woman delivers her first child at home rather than at a hospital or clinic (column 1), education does increase the probability that a woman had a doctor or midwife assist in the delivery. According to the IV estimate in column 2, each year of schooling increases the likelihood of a trained health practitioner present at the birth of the first child by 29 percentage points. This is a large effect considering that only 48 percent of women on average have a trained professional assist with the birth of their first child.

Women's schooling also increases the likelihood that children are immunized against preventable diseases. The WHO considers children to be fully immunized when they have received vaccines for both tuberculosis and the measles along with three doses each of the polio vaccine and the DPT (diphtheria, pertussis, and tetanus) vaccine by the end of their first year. As shown in columns 4-7, women's education increases the likelihood that first-born children (who have reached 1 year of age) are fully immunized. Each additional year of schooling increases the probability the first-born child received the tuberculosis vaccine by 11 percentage points, the measles vaccine by 15 percentage points, the complete polio vaccine by 13 percentage points, and the DPT vaccine by 12 percentage points. Moreover, higher mother's education also increases the chances that

in the youngest cohort (born in 1990) are 21 years old, and therefore are unable to report on births at age 22. It is possible, of course, to expand the range by excluding these cohorts and shrinking the bandwidth. Results (available upon request) are not significantly altered after making this adjustment.

a first-born child receives a vitamin A supplement (column 8). Periodic dosing with this supplement prevents severe vitamin A deficiency, which can lead to blindness and also exacerbate common illnesses such as diarrhea and the measles.

Table 6 presents impacts of female schooling on indicators of general health for first-born children. I find large and statistically significant impacts of education on child height. Controlling for age, first born children of women with additional schooling are .39 standard deviations taller. Relatedly, they are also 13 percentage points (37 percent) less likely to be stunted, a measure of the long term effects of malnutrition and chronic illness.[20] In line with this, these children are also 4.2 percentage points (or 7 percent) less likely to be anemic (column 5), a condition of low levels of hemoglobin in the blood brought about mainly by poor nutrition and, among children under 5, by prolonged exposure to malaria.

However, there is no difference in the probability that the first-born child died before reaching 1 year of age. This finding is in contrast to Chou et al. (2010), who find a significant reduction in child mortality by mother's education in Taiwan, and Breierova and Duflo (2004), who find that increased average household education (both mother and father) reduces child mortality in Indonesia.

There are three possible explanations for this discrepancy. First, although schooling does increase investments in nutrition and vaccinations, just 10 percent of under-5 deaths were attributable to malnutrition or diseases that can be prevented with a vaccine (Uganda Bureau of Statistics (UBOS) et al. (2008)).[21] Education does not appear to have an effect on child weight (controlling for age) or on the probability that a first-born child shows signs of wasting, which is an

[20]A child is defined as stunted if his height-for-age Z-score is less than -2 standard deviations from the median using the CDC Standard Deviation-derived Growth Reference Curves from the NCHS/FELS/CDC Reference Population. See Uganda Demographic and Health Surveys 2006 Final Report for details.

[21]The most common cause of under-5 mortality is malaria (31.9 percent) followed by perinatal and early neonatal conditions (18.1 percent) (Uganda Bureau of Statistics, 2008).

indicator of acute malnutrition episodes or illness that caused weight loss in the period immediately prior to measurement. Indeed, mothers on both sides of the cutoff were equally likely to report their first-born child had fever or diarrhea in the two weeks prior to the survey (results not shown). Second, this study is limited to self-reported child mortality information from a relatively small sample which may be more likely to suffer from measurement error, whereas Chou et al. (2010) relies on administrative census data on all child deaths and Breierova and Duflo (2004) have a sample 20 times larger than the present study. Finally, as described in fuller detail in Section 5.3, there is evidence that women in the child health sample are negatively selected on education overall (but not differentially at the cutoff). Thus, they are likely to come from more disadvantaged households, to have lower ability, or be negatively selected on some other unobservable as well, which may tend to bias estimates of treatment effects downward.

Overall, these results suggest that mothers' education increases children's long-term health chances, but has less impact on more acute health outcomes that effect early child mortality (again keeping in mind that results for the first-born child subsample are likely biased downward due to negative selection overall).

5.4. Mechanisms and other downstream impacts

A multitude of potential channels have been suggested that connect female education to fertility and child outcomes. These range from changes in opportunity costs as in neoclassical models (Becker and Lewis (1973) and Willis (1973)) or the effects of positive assortative mating (Behrman and Rosenzweig (2002)), to knowledge acquisition and, in particular, learning about family planning or health (Glewwe (1999), Grossman (2006), and Rosenzweig and Schultz (1989)), to changes in women's conceptions with respect to their role in the family or society (Lavy and Zablotsky (2015)) or to their bargaining power (Thomas (1990) and Jejeebhoy et al. (1995)). While the DHS contain data a number of these dimensions, most are measured contemporaneously to the survey date and are thus potentially consequences of lower fertility or improved child health rather than causes of them. For example, women in Uganda may be more likely to have better employment outcomes in their 20s as a result of starting families later or having fewer children rather than the reverse.

In this paper, I therefore focus on potential mechanisms that were plausibly fixed before women began having children. These include the onset of sexual activity, contraceptive use before first birth, marriage timing, and husband education. In Uganda, both the onset of sexual activity and marriage take place largely in the adolescent years. The mean age at first sexual intercourse in the sample is 16, and the mean age at first marriage is 17. By age 20, 86 percent of women have had sex and 73 percent are married. Results are presented in Table 7.

I find that while women who get more schooling are just as likely to have sex as their peers at young ages (columns 1 and 2), they also delay marriage (columns 3 and 4). Schooling reduces the probability of marriage by age 15 by more than 14 percent and by age 20 by almost 5 percent. Similarly, education has a large impact on early contraceptive use. As shown in column 6, an additional year of schooling increases the likelihood of contraceptive use before the birth of a first child by 36 percent (an increase of 6.5 percentage points off a base of 18 percent). Together, these results rule out a possible "incarceration effect" and suggest instead that schooling provides women with both the desire and ability to reduce early pregnancies without resorting to abstinence only.

Finally, I explore whether there is any evidence of positive assortative mating. Because the reform essentially shifted the entire education distribution among affected cohorts upward, this raises the possibility that at least some treated women looked for more educated spouses from among the pool of men who would have otherwise, in the absence of UPE, married unaffected women to the left of the cutoff. In this case, treatment effects would become uninterpretable since they would rely on both treated women having more educated husbands and untreated women having relatively less educated husbands, a violation of the monotonicity assumption. However, I find no evidence of assortative mating of any kind. Column 5 shows that women with more schooling who do marry match with men of similar schooling levels as their less educated peers.[22] Thus it appears that husband education plays no role in the effects measured in this paper.

There are other downstream impacts on labor market outcomes and household wealth, but it is less clear how they should be interpreted. Table 8 shows that schooling increases the probability a woman worked in the last year by 9.2 percent, that this was employed work (rather than an own enterprise or family business) by 33 percent, and that they were paid in cash (rather than in kind or not at all) by 13 percent. Additional female schooling also increases scores on a standardized index of household assets and quality by .12 standard deviations.[23] Recall that there were no differences in household wealth between treated and untreated women when the sample cohorts were still of school age in the 1995 and 2000/01 DHS (Figure A8), supporting the notion that the gains in wealth are ultimately the result of additional schooling.

Improved labor market outcomes are a possible mechanism through which women's education induces a trade-off between child quantity and child quality. Yet it is also possible these gains come at the expense of worse labor market outcomes for women in cohorts just to the left of the cutoff. If so, the estimated impacts of female schooling on fertility and child health measured in this paper may, again, be biased upward. At the same time, because these outcomes are measured at the time of survey – and not, typically, around the time women begin childbearing – it is unclear if changes in employment opportunities are really driving the fertility and first-born child results rather than the other way around. For example, it could be the case that women's fertility choices

[22] The sample size is lower than in columns 1-4, despite using the same data, because (a) not all women are married, and (b) the data is restricted to women who are still married to their first husbands. There is no differential effect on dissolution of first marriages or polygamy (results available upon request).

[23] The index includes assets such as televisions, radios, and refrigerators, as well as the type of construction used in the home (roofing and flooring), and access to water sanitation.

are driven by factors unrelated to employment considerations (e.g. Lavy and Zablotsky (2015)), but then, finding themselves with fewer children, women are less constrained to participate in the labor market. Because of this data limitation, I am unable to further explore the relative timing of fertility decisions and labor market gains to be able to drawn any conclusions about their role as a potential mechanism.

5.5. Education or income effects?

A potential concern with interpreting the results in this paper as the effects of increased schooling is that there are women in affected cohorts who would have attended school in the absence of the UPE reform, and for whom therefore the program served principally as a household income transfer. Decreases in fertility and improvements in child outcomes may be driven wholly, or in part, by income effects to these women rather than by education effects alone. To shed light on this, I implement a difference-in-differences-in-differences empirical strategy.

The first difference compares women aged 15-17 before and after the 1997 reform (using the 1995 DHS and the 2000 DHS). Women aged 15-17 in 2000 (who were 12-14 in 1997) benefitted from the introduction of free primary schooling, while those aged 15-17 in 1995 did not. Thus this first difference captures the effect of the reform and any time trend in outcomes. The second difference compares women aged 18-20 before and after the reform. These women were all too old to benefit from UPE and thus serve as a plausible control for any trends over time in outcomes between 1995 and 2000. The third difference compares outcomes for each age group and time period by household wealth levels, proxied by whether a woman's home has a dirt floor. Given the young ages of women, home flooring material is likely highly correlated with family wealth. Nearly 90 percent 15-17 year olds, and half of 18-20 year olds, are dependents (i.e. neither a household head or a spouse) in another adult's household. Approximately 65 percent of women aged 15-20 live in households with dirt floors. In the pre-reform 1995 DHS data, women from homes with dirt floors obtained 4 years of schooling on average, while women from homes with finished floors completed 7 years of schooling on average (i.e. finished primary school). Thus the household flooring measure separates women who likely would have completed primary school without UPE from those who would likely stop schooling without the removal of school fees. I estimate the following regression equation:

$$\begin{aligned} Y_i = \beta_0 + \beta_1(young_i \times post_i) + \beta_2(young_i \times poor_i \times post_i) + \beta_3(poor_i \times post_i) + \\ \beta_4 post_i + \beta_5(young_i \times poor_i) + \beta_6 young_i + \beta_7 poor + \varepsilon_i \end{aligned} \tag{4}$$

where *young* is an indicator for whether a woman was aged 15-17 rather than aged 18-20, *poor* is an indicator for whether a woman's household has a dirt floor, and $post = 1$ if the survey period is the 2000 DHS and zero otherwise. The main coefficients of interest are β_1 and β_2, which measure the impact of UPE on affected women from relatively well-off households and the differential impact of the reform on affected women from relatively poorer households, respectively. Results are presented in Table 9.

Column 1 estimates the effect of UPE on educational attainment. The coefficient on β_1, the effect of UPE on relatively richer affected women, is small and not statistically different from zero. This is consistent with the fact that richer women already achieved completed primary schooling prior to the reform. On the other hand, the coefficient on β_2, the differential effect of UPE on relatively poorer affected women, is large (nearly .90 years additional schooling) and statistically significant. Again, this is as expected: women from poorer households were potentially constrained from completing additional schooling by school fees prior to 1997. Together, these two results provide justification for the concern that the reform also served as a household income transfer to a subset of women in the affected cohorts.

Columns 2-5 of Table 9 explore whether, and to what extent, the fertility results are being driven by women from richer households who benefit from this income transfer rather than by women from poorer households who obtain more schooling. In particular, I estimate equation (4) substituting for the dependent variable the likelihood a woman has become sexually active, the likelihood of marriage, the probability a woman has begun childbearing, and the number of children a woman has. A clear pattern emerges. Free primary education appears to have had no effect for affected women from relatively wealthier households. In contrast, affected women from poorer households are statistically significantly less likely to be sexually active, married, or have children. Additionally, although measured less precisely, affected women from poorer households have fewer children overall. These results strongly suggest that the main fertility results found in this paper are indeed being driven by changes in women's schooling rather than by household income effects accruing to women who would have attended primary school in the absence of UPE. Although I am not able to conduct a similar test for child health investments and outcomes (because few women aged 15-17 have borne children), we know that the sample of women used for the first-born child results in section 5.4 is negatively selected overall on education. Thus, for these women the reform likely served primarily to remove a barrier (school fees) to continued schooling and not as a household income transfer.

6. Conclusion

Given the strong observed correlations between female education and fertility and child health outcomes, development policy-makers have long focused on women's education as a means to accomplish multiple development goals. Indeed, economic theory predicts women with more education will trade child quantity for child quality. Yet while empirical studies have consistently found that schooling delays fertility, with mixed evidence of an effect on total fertility, and that in some contexts education improves child health, there is no direct evidence of such a quantity-quality trade-off.

The results in this paper bear out both predictions. I find that women's education not only reduces fertility during adolescence, it also reduces overall fertility years after schooling is complete. Moreover, in line with theory, I find that women with more education trade off additional fertility for increased investments in the children they do have. In particular, children born to women with more education are more likely to be protected against common diseases and are better nourished. In terms of mechanisms, in addition to finding evidence of delayed marriages, this paper also adds to the literature that finds that female schooling increases early contraceptive use.

Finally, this study contributes new evidence of the effects of reforms that remove school fees, a policy change that more than a dozen Sub-Sahara African countries have implemented in recent years. In particular, I find that eliminating primary school fees also increases secondary school transition and completion rates even when secondary school fees remain in place. This suggests that the presence of these fees constrains even high ability students (particularly girls) from achieving higher levels of education. Additionally, the evidence shows that schooling has long-run impacts on labor market outcomes and household wealth.

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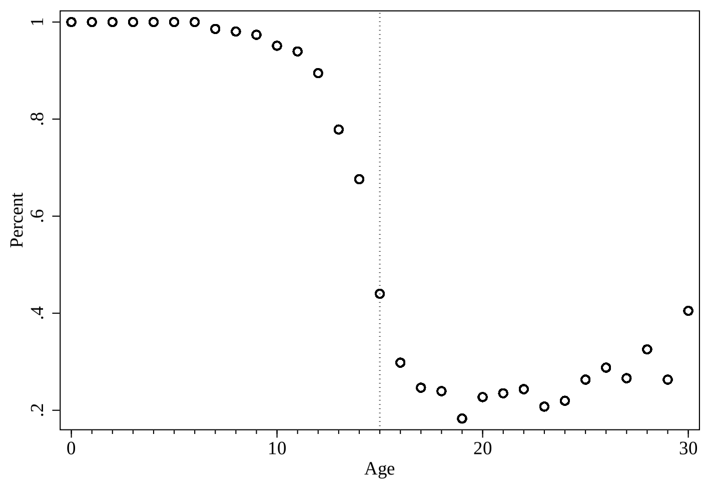
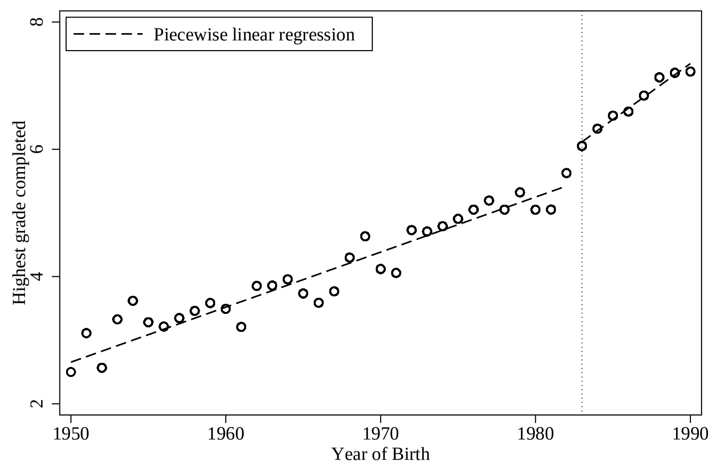
Figures and Tables

Figure 1: Fraction with incomplete primary education prior to reform



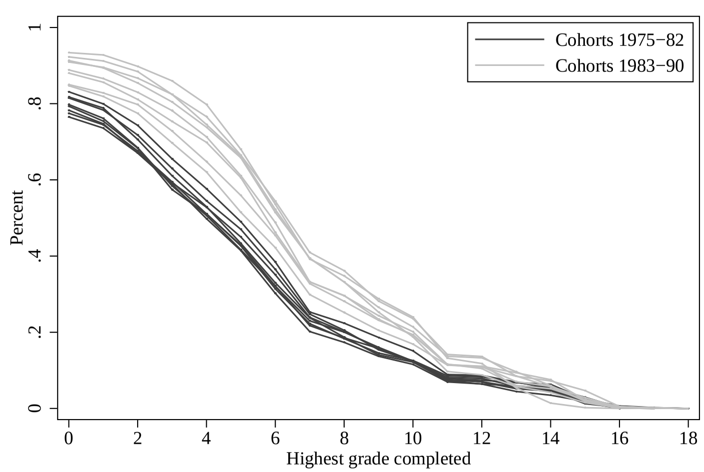
Notes: Data from DHS (1995) Female Survey.

Figure 2: Average highest grade completed by birth cohort



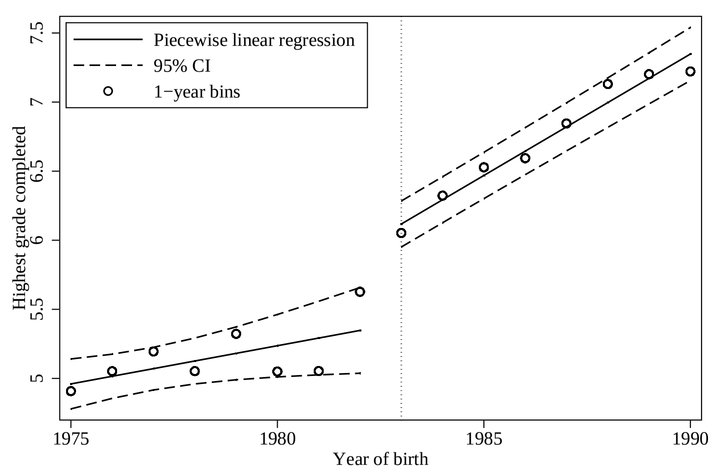
Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Female Surveys. Sample restricted to cohorts born 1950−1990.

Figure 3: Reverse CDF of education by birth cohort



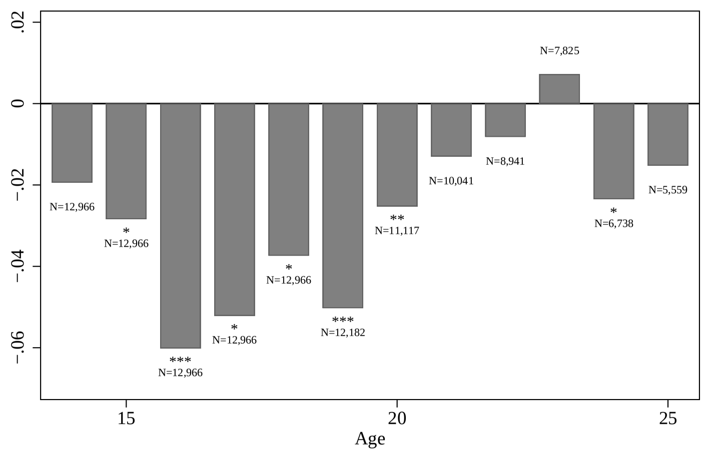
Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Female Surveys. Sample restricted to cohorts born 1975−1990.

Figure 4: First stage: Year of birth and Schooling



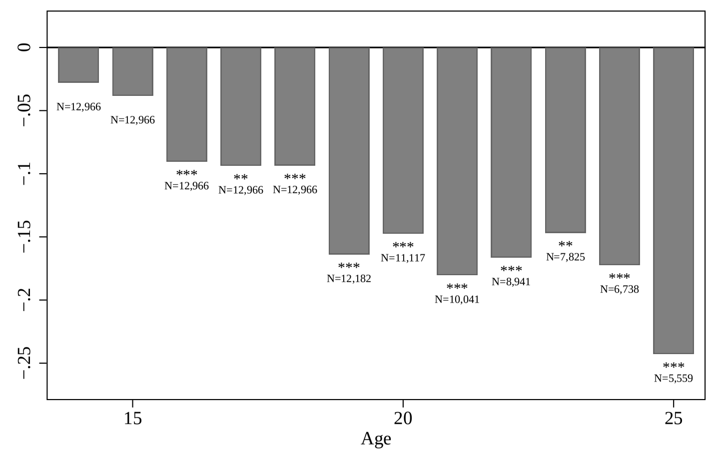
Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Female Surveys. Sample restricted to cohorts born 1975−1990.

Figure 5: Differences in probability of first birth by age at the discontinuity



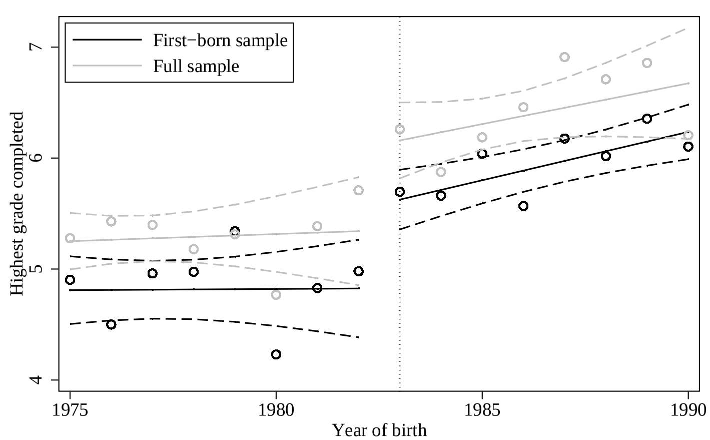
*Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Female Surveys. Sample restricted to cohorts born 1975−1990. Each bar plots the coefficient on education from a separate regression. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Figure 6: Differences in number of births by age at the discontinuity



*Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Female Surveys. Sample restricted to cohorts born 1975−1990. Each bar plots the coefficient on education from a separate regression. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Figure 7: First-born sample first stage



Notes: Data from DHS (1995, 2000/01, 2006, and 2011) Female Surveys. Full sample includes women aged 15−26 at time of survey from cohorts born 1975−1990. First−born sample includes women aged 15−26 at time of survey from cohorts born 1975−1990 who had their first child in the 5 years preceding the survey. Dashed lines represent 95% confidence intervals.

Table 1: Sample Characteristics

	2000/01 DHS	2006 DHS	2009 MIS DHS	2011 DHS
Cohorts included	1975-1981	1975-1987	1975-1990	1975-1990
Age at cutoff (born 1983)	n/a	23	26	28
Years of schooling	5.63	5.50	5.81	6.19
	[4.07]	[4.15]	[4.24]	[4.49]
Complete primary school	0.39	0.37	0.40	0.44
Complete secondary school	0.07	0.08	0.08	0.12
Able to read sentence	0.56	0.50		0.51
Worked in last year	0.74	0.86	0.70	0.79
Wealth index	0.17	0.18	0.14	0.11
	[1.09]	[1.16]	[1.07]	[1.06]
Ever married	0.81	0.82		0.88
Age at first marriage	16.89	17.27		17.83
	[2.51]	[2.86]		[3.52]
Husband's years of schooling	7.16	6.89		7.49
	[4.09]	[4.05]		[4.64]
Ever sexually active	0.93	0.93		0.96
Age at first sex	16.19	16.25		16.70
	[2.16]	[2.38]		[2.89]
Ever pregnant	0.79	0.83	0.85	0.89
Age at first birth	17.71	18.08	17.95	18.44
	[2.29]	[2.65]	[3.10]	[3.20]
Number of children born	1.80	2.56	2.87	3.32
	[1.39]	[1.98]	[2.16]	[2.27]
Number of observations	2183	3984	2318	4481

Note: each sample restricted to woman 19 or older at the time of survey and cohorts born 1975-1990. Standard deviations in brackets.

Table 2: Education outcomes

	Highest grade completed (1)	Primary completed (2)	Some secondary (3)	Secondary completed (4)	Can read sentence (5)
Year of birth $\geq$ 1983	0.715 ***	0.057 ***	0.048 ***	0.028 ***	0.039 ***
	(0.159)	(0.013)	(0.003)	(0.007)	(0.011)
Year of birth	0.055 **	0.009 ***	0.005 ***	0.000	0.003 ***
	(0.026)	(0.002)	(0.001)	(0.001)	(0.001)
(Year of birth $\geq$ 1983) $\times$	0.120 ***	0.009 ***	0.011 ***	0.002	0.007 ***
Year of birth	(0.029)	(0.002)	(0.002)	(0.002)	(0.002)
Mean of dep. var.	5.82	0.40	0.29	0.09	0.52
Std. dev.	[4.28]	[0.49]	[0.45]	[0.29]	[0.50]
Number of observations	12966	12966	12966	12966	10328

*Note: Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. In each regression the sample is restricted to individuals born between 1975-1990. Data for columns 1-4 are from the 2000/01, 2006, and 2011 DHS and the 2009 MIS DHS; data for column 5 is from the 2000/01, 2006, and 2011 DHS. Standard errors, clustered at the survey cluster and year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.*

Table 3: Total fertility

	Total children born by: 2006 (1)	2009 (2)	2011 (3)	Mean birth interval (months) (4)
Highest grade completed	-0.092 ***	-0.366 ***	-0.520	1.360 ***
	(0.038)	(0.100)	(0.636)	(0.512)
Year of birth	-0.351 ***	-0.236 ***	-0.190	-0.230 ***
	(0.011)	(0.029)	(0.117)	(0.046)
(Year of birth $\geq$ 1983) $\times$ Year of birth	0.015	-0.019	-0.051	-0.294 ***
	(0.011)	(0.022)	(0.039)	(0.117)
Mean of dep. var.	2.56	2.87	3.32	31.46
Std. dev.	[1.98]	[2.16]	[2.27]	[12.98]
Number of observations	3984	2318	4481	8774
First-stage effect on education	0.895 ***	1.444 ***	0.276	0.707 ***
	(0.148)	(0.420)	(0.339)	(0.220)

*Note: Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. Columns 1-3 use data from the 2006 DHS, the 2009 MIS DHS, and the 2011 DHS, respectively. Column 4 restricts the sample to women with multiple births and uses data from the 2000/01, 2006 and 2011 DHS and 2009 MIS DHS combined. In each regression the sample is restricted to individuals born between 1975-1990. Standard errors, clustered at the survey cluster and year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Table 4: Sample for child outcomes

	Highest grade completed (1)	Had first born between 15-21 years old (2)	First born 15-21: Highest grade completed (3)	In child sample (4)	In child sample: Highest grade completed (5)
Year of birth $\geq$ 1983	0.643 ***	0.019	0.709 ***	-0.022	0.799 ***
	(0.218)	(0.013)	(0.153)	(0.038)	(0.268)
Year of birth	0.062 *	-0.007 ***	0.022	-0.002	0.002
	(0.037)	(0.002)	(0.025)	(0.003)	(0.037)
(Year of birth $\geq$ 1983) $\times$	0.166 ***	0.001	0.146 ***	0.001	0.085 *
Year of birth	(0.046)	(0.004)	(0.028)	(0.008)	(0.048)
Mean of dep. var.	5.72	0.71	4.90	0.31	5.37
Std. dev.	[4.42]	[0.46]	[3.68]	[0.46]	[3.40]
Number of observations	8262	8262	5834	11158	3431

Note: *Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. Columns 1-8 use data from the 1995, 2000/01, 2006 and 2011 DHS. Data restricted to women aged 15-26 at the time of the survey. Columns 4-8 restricted to women whose children are at least 1 year old. In each regression the sample is restricted to individuals born between 1975-1990. Standard errors, clustered at the survey cluster and year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Table 5: First-born children – health inputs

	Childbirth at home	Trained assistant at birth	Breastfed at least 6 months	Vaccines				Vitamin A supplement
				Tuberculosis	Measles	Polio	Diptheria	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Highest grade completed	0.063	0.290 ***	0.016	0.106 ***	0.116 ***	0.077 ***	0.080 ***	0.064 ***
	(0.061)	(0.077)	(0.020)	(0.026)	(0.047)	(0.033)	(0.033)	(0.025)
Year of birth	-0.004	-0.023 ***	0.003	-0.004	-0.002	-0.005	-0.011 ***	0.057 ***
	(0.006)	(0.008)	(0.003)	(0.005)	(0.006)	(0.006)	(0.003)	(0.004)
(Year of birth $\geq$ 1983) $\times$	-0.024 ***	0.031 ***	-0.008	-0.002	0.001	0.000	0.027 ***	-0.076 ***
Year of birth	(0.008)	(0.012)	(0.007)	(0.007)	(0.012)	(0.007)	(0.007)	(0.009)
Mean of dep. var.	0.40	0.48	0.91	0.91	0.77	0.59	0.64	0.56
Std. dev.	[0.49]	[0.50]	[0.29]	[0.29]	[0.42]	[0.49]	[0.48]	[0.50]
Number of observations	3422	3425	2562	2367	2365	2364	2394	2593
First-stage effect on education	0.809 ***	0.808 ***	0.784 **	0.931 ***	0.930 ***	0.912 ***	0.878 ***	1.423 ***
	(0.279)	(0.278)	(0.357)	(0.366)	(0.365)	(0.363)	(0.349)	(0.416)

Note: *Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. Columns 1-8 use data from the 1995, 2000/01, 2006 and 2011 DHS. Data restricted to women aged 15-26 at the time of the survey. Each regression controls for child age at time of survey. In each regression the sample is restricted to individuals born between 1975-1990. Standard errors, clustered at the survey cluster and year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Table 6: First-born children – health outcomes

	Height for age (1)	Stunting (2)	Weight for age (3)	Wasting (4)	Anemia (5)	Infant mortality (6)
Highest grade completed	0.385 ***	-0.129 ***	0.070	0.001	-0.042 ***	0.026
	(0.093)	(0.041)	(0.096)	(0.023)	(0.016)	(0.027)
Year of birth	-0.005	0.006	0.004	0.000	0.009	-0.007 **
	(0.012)	(0.005)	(0.009)	(0.003)	(0.009)	(0.004)
(Year of birth $\geq$ 1983) $\times$	-0.041	0.005	-0.037	0.000	-0.031 **	-0.002
Year of birth	(0.032)	(0.015)	(0.034)	(0.008)	(0.014)	(0.003)
Mean of dep. var.	-1.54	0.35	-1.08	0.20	0.62	0.10
Std. dev.	[1.33]	[0.48]	[1.16]	[0.40]	[0.48]	[0.30]
Number of observations	1524	1524	1524	1524	1135	2624
First-stage effect on education	1.315 ***	1.315 ***	1.315 ***	1.315 ***	2.050 ***	0.923 ***
	(0.391)	(0.391)	(0.391)	(0.391)	(0.506)	(0.343)

*Note: Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. Data are from the 1995, 2000/01, 2006 and 2011 DHS. Data restricted to women aged 15-26 at the time of the survey. Each regression controls for child age at time of survey. Column 6 restricts data to children born at least 1 year prior to the survey. In each regression the sample is restricted to individuals born between 1975-1990. Standard errors, clustered at the survey cluster and year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Table 7: Early reproductive behaviors and marriage outcomes

	Sexually active by age 15 (1)	Sexually active by age 20 (2)	Married by age 15 (3)	Married by age 20 (4)	Husband education (5)	Contraceptive use before first birth (6)
Highest grade completed	-0.002	-0.009	-0.034 ***	-0.035 *	-0.360	0.065 ***
	(0.016)	(0.011)	(0.015)	(0.020)	(0.595)	(0.021)
Year of birth	-0.008 ***	-0.002	-0.001	-0.002	0.025	0.013 ***
	(0.003)	(0.002)	(0.002)	(0.003)	(0.059)	(0.004)
(Year of birth $\geq$ 1983) $\times$	-0.005	-0.003 *	-0.004	-0.004	0.163 ***	0.003
Year of birth	(0.004)	(0.002)	(0.004)	(0.003)	(0.066)	(0.006)
Mean of dep. var.	0.35	0.91	0.23	0.73	6.81	0.18
Std. dev.	[0.48]	[0.29]	[0.42]	[0.44]	[4.08]	[0.38]
Number of observations	10160	8739	10648	9199	5346	5727
First-stage effect on education	0.581 ***	0.648 ***	0.576 ***	0.642 ***	0.387 *	1.072 ***
	(0.191)	(0.230)	(0.176)	(0.227)	(0.234)	(0.256)

*Note: Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. Columns 1-5 use data from the 2000/01, 2006 and 2011 DHS combined. Column 6 uses data from the 2000/01, and 2006 DHS combined. In each regression the sample is restricted to individuals born between 1975-1990. Standard errors, clustered at the survey cluster and year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Table 8: Labor market and household wealth

	Working	Employed by other	Paid in cash	Wealth index
	(1)	(2)	(3)	(4)
Highest grade completed	0.073 ***	0.056 ***	0.048 **	0.116 ***
	(0.031)	(0.016)	(0.024)	(0.034)
Year of birth	-0.014 ***	-0.004 ***	-0.004 *	-0.011 *
	(0.005)	(0.002)	(0.002)	(0.006)
(Year of birth $\geq$ 1983) $\times$	-0.029 ***	0.005	0.006	-0.005
Year of birth	(0.005)	(0.003)	(0.006)	(0.008)
Mean of dep. var.	0.79	0.17	0.37	0.15
Std. dev.	[0.41]	[0.37]	[0.48]	[1.10]
Number of observations	12958	8561	10158	12966
First-stage effect on education	0.711 ***	0.696 ***	0.861 ***	0.715 ***
	(0.163)	(0.236)	(0.227)	(0.164)

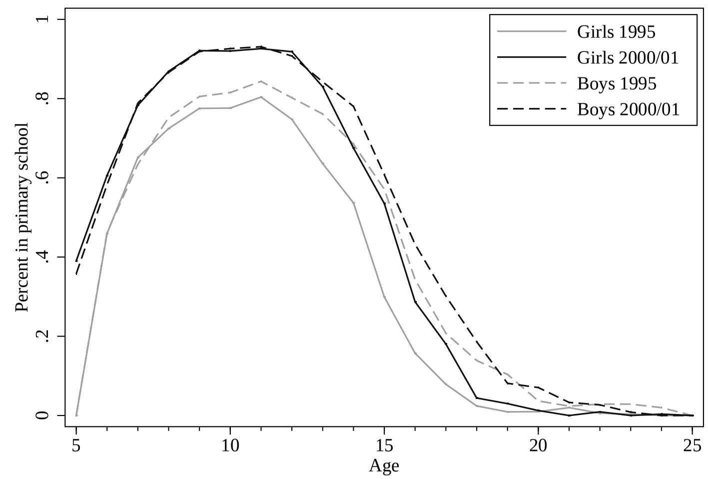
*Note: Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. Columns 1, 3, and 4 use data from the 2000/01, 2006, and 2011 DHS, and the 2009 MIS DHS combined. Column 2 use data from the 2000/01, 2006 and 2011 DHS combined. In each regression the sample is restricted to individuals born between 1975-1990. Standard errors, clustered at the survey cluster and year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Table 9: Heterogenous impacts by household wealth

	Highest grade completed (1)	Sexually active (2)	Married (3)	Has any children (4)	Number of children (5)
Age 15-17 × post (β_1)	-0.029	0.048	-0.028	-0.029	-0.060
	(0.463)	(0.119)	(0.116)	(0.114)	(0.223)
Age 15-17 × post × poor (β_2)	0.886 **	-0.174 ***	-0.133 ***	-0.064 *	-0.128
	(0.427)	(0.054)	(0.039)	(0.034)	(0.085)
Poor × post	-0.161	0.002	0.005	0.006	0.081
	(0.347)	(0.030)	(0.025)	(0.018)	(0.080)
Post	0.537 *	0.009	-0.010	0.014	0.027
	(0.305)	(0.072)	(0.090)	(0.095)	(0.210)
Age 15-17 × poor	0.526	-0.013	-0.060 *	-0.136 ***	-0.290 ***
	(0.346)	(0.054)	(0.032)	(0.031)	(0.059)
Age 15-17	-0.762 ***	-0.400 ***	-0.351 ***	-0.334 ***	-0.519 ***
	(0.296)	(0.074)	(0.061)	(0.064)	(0.134)
Poor	-3.227 ***	0.102 ***	0.261 ***	0.237 ***	0.386 ***
	(0.300)	(0.030)	(0.022)	(0.017)	(0.053)
Constant	7.188 ***	0.729 ***	0.473 ***	0.449 ***	0.661 ***
	(0.229)	(0.049)	(0.043)	(0.057)	(0.128)
Number of observations	3708	3665	3708	3708	3708

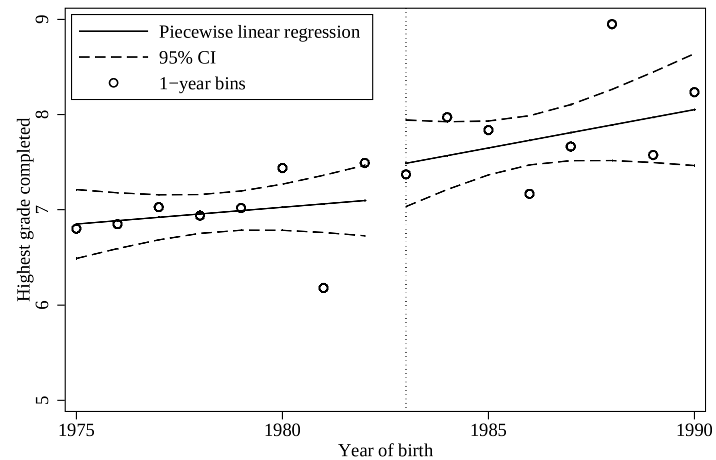
*Note: Data from 1995 and 2000/01 DHS. Standard errors, clustered at the year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Figure A1: Primary school attendance by age - 1995 and 2000/01



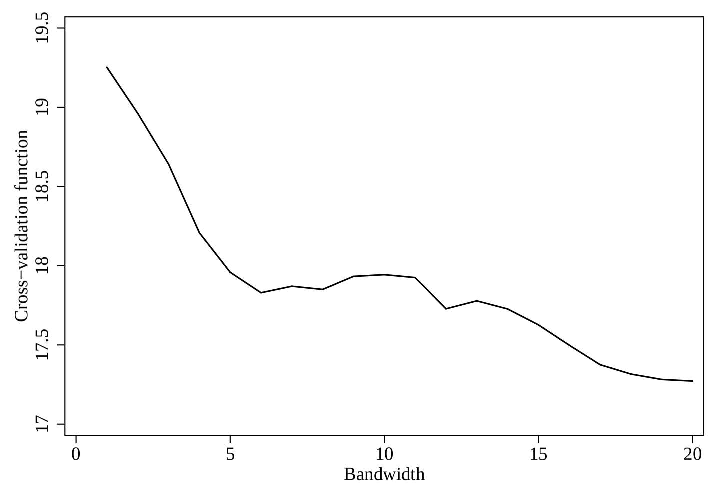
Notes: Data from DHS (1995 and 2000/01) Household Surveys.

Figure A2: Avg. highest grade completed by birth cohort - Males



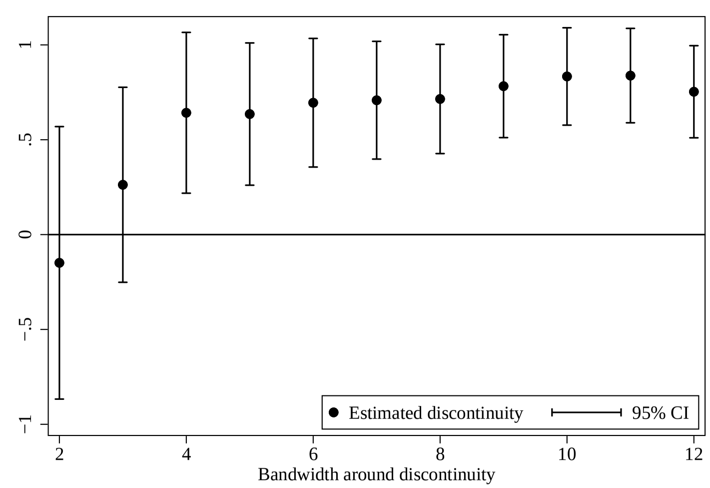
Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Male Surveys. Sample restricted to cohorts born 1975−1990.

Figure A3: Cross-validation function using year of birth



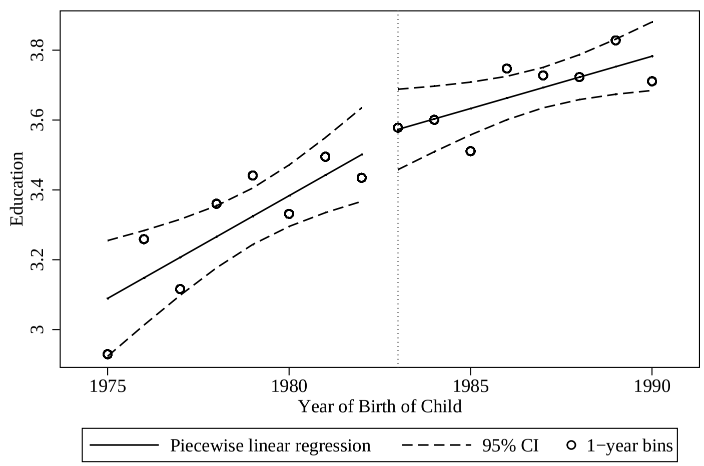
Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Female Surveys.

Figure A4: Discontinuity as function of bandwidth (window size)



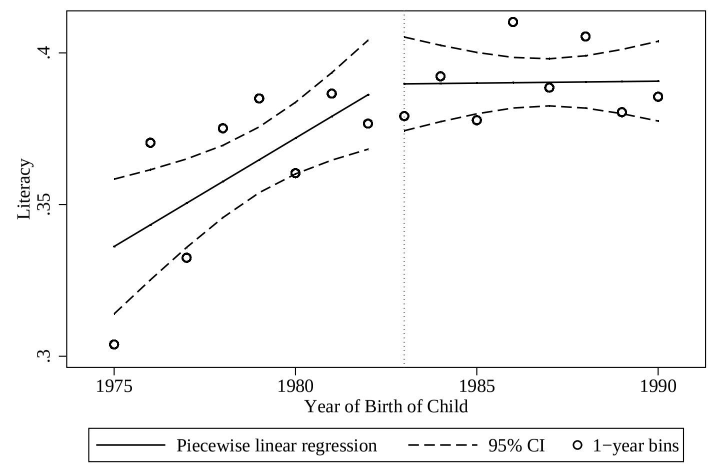
Notes: Data from DHS (2000/01, 2006 and 2011) and DHS MIS (2009) Female Surveys.

Figure A5: Education - mothers of cohort women



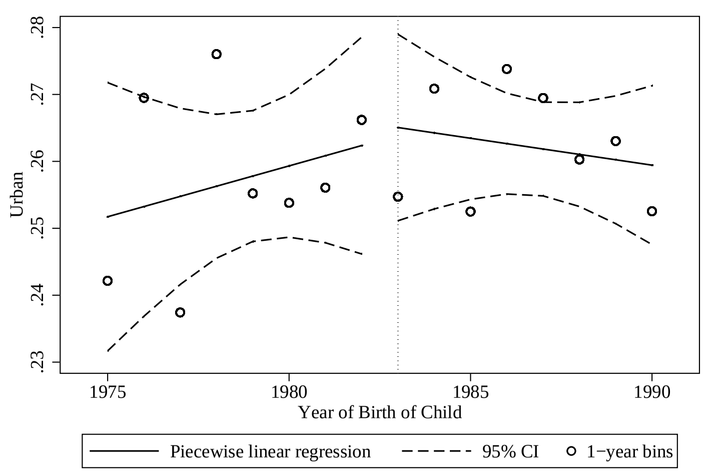
Notes: Data from DHS (1995 and 2000/01) Female Surveys. Sample restricted to women who had a child between 1975−1990.

Figure A6: Literacy - mothers of cohort women



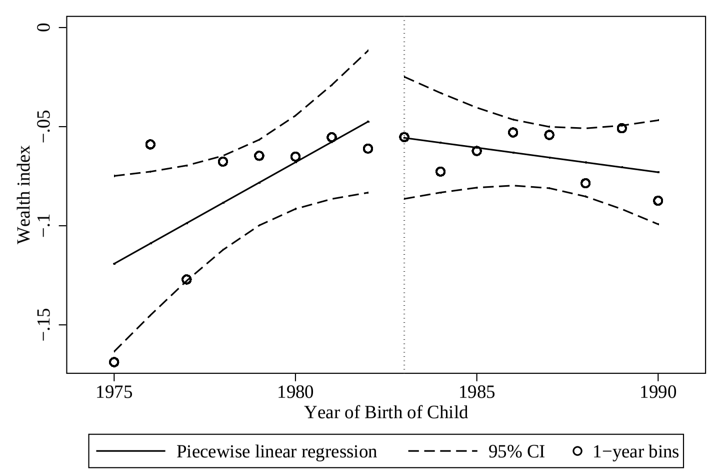
Notes: Data from DHS (1995 and 2000/01) Female Surveys. Sample restricted to women who had a child between 1975−1990.

Figure A7: Urban - mothers of cohort women



Notes: Data from DHS (1995 and 2000/01) Female Surveys. Sample restricted to women who had a child between 1975−1990.

Figure A8: Wealth Index - mothers of cohort women



Notes: Data from DHS (1995 and 2000/01) Female Surveys. Sample restricted to women who had a child between 1975−1990.

Table A1: Evidence of discontinuity in other data sets

Data source:	2002 Census		2008 National Service Delivery Survey		2012 Living Standards Measurement Survey	
Dependent variable:	Primary completed		Highest grade completed		Highest grade completed	
	Females	Males	Females	Males	Females	Males
	(1)	(2)	(3)	(4)	(5)	(6)
Year of birth $\geq$ 1983	0.043 ***	0.014 ***	0.433 **	-0.047	0.681 *	-0.039
	(0.003)	(0.003)	(0.199)	(0.197)	(0.404)	(0.430)
Year of birth	0.008 ***	0.004 ***	0.105 ***	0.093 ***	0.080	0.221 ***
	(0.000)	(0.000)	(0.035)	(0.033)	(0.064)	(0.068)
(Year of birth $\geq$ 1983) $\times$	0.001	-0.047 ***	-0.064	-0.095 **	0.167 **	-0.106
Year of birth	(0.002)	(0.002)	(0.047)	(0.046)	(0.085)	(0.088)
Mean of dep. var.	0.33	0.48	6.78	7.60	6.40	7.98
Std. dev.	[0.47]	[0.50]	[3.26]	[3.22]	[3.95]	[3.71]
Number of observations	452269	434924	4577	4332	1552	1342

Note: *Years of birth has been re-centered at the discontinuity (1983) so that the coefficient on the discontinuity may be interpreted directly. In each regression the sample is restricted to individuals born between 1975-1990. Standard errors, clustered at the year of birth level, are in parentheses. ***, **, and * indicate significance at the 1, 5, and 10 percent levels respectively.*

Highlights

- This paper measures the impact of female schooling on fertility and child health.
- Identification exploits an age discontinuity in exposure to free primary education.
- Schooling delays and decreases fertility, and increases child health investments.
- Mechanisms include delayed marriage and use of contraceptives before first birth.
- Schooling has downstream impacts on labor market outcomes and household wealth.